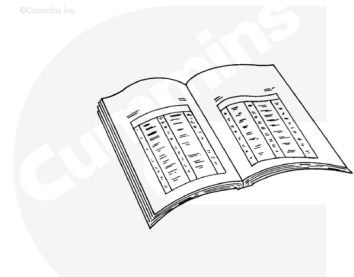


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Drain the engine oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Remove the brackets for the marine gear oil cooler plumbing, if equipped. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)

Remove

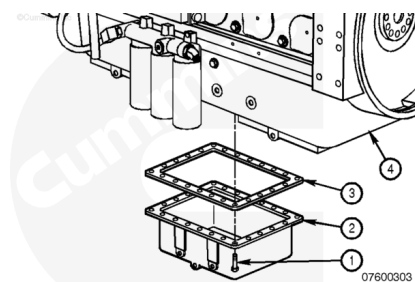
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Note : The K38, K50, QSK38, and QSK50 engines use a variety of lubricating oil pan and cover plate configurations.

Remove the capscrews (1), oil pan (2), and gasket (3) from the oil pan adapter (4).

Inspect gaskets for reuse (3).



Clean and Inspect for Reuse

⚠ WARNING ⚠

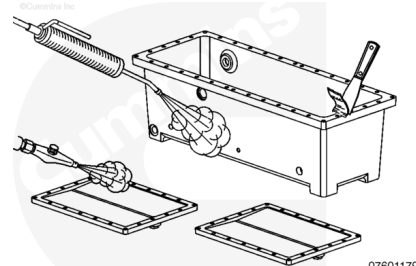
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

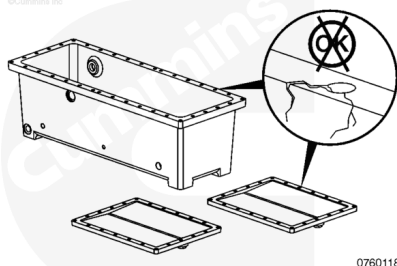


Clean all gasket surfaces on the oil pan.

Steam clean all parts and dry with compressed air.

Use solvent to clean drain plug magnets and dry with compressed air.

Inspect all areas of the oil pan for cracks or other damage. If any of the areas are cracked or otherwise damaged, the oil pan **must** be replaced.



07601180

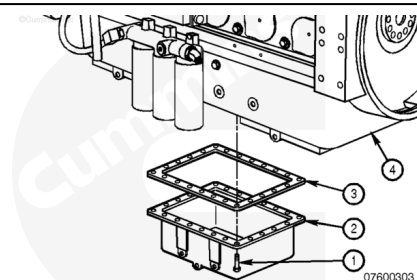
Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Do not use gasket cement on the oil pan gaskets. Gasket cement will prevent the gasket from sealing properly.



07600303

Position the oil pan (2) and gaskets (3) on the oil adapter (4) with capscrews.

Note : Shown is the tightening pattern for the cover plate. The same tightening pattern and value is used for the oil pan. The number of capscrews being tightened can vary due to application. Use the pattern shown as a guide when tightening the capscrews.

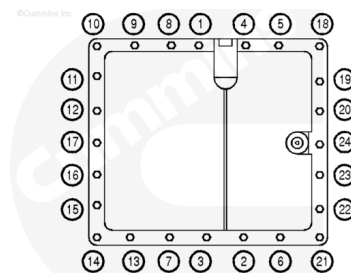
Torque Value:

Cover Plate Capscrew 45 n•m [33 ft-lb]

Install and tighten the drain plug(s).

Torque Value:

Oil Pan Drain Plug 60 n•m [45 ft-lb]



07600350

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

- Install the brackets for the marine gear oil cooler plumbing, if equipped. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)
- Install new lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Fill the engine with clean 15W-40 oil. Refer to Procedure 018-017 in Section V for oil pan capacities. (</qs3/pubsys2/xml/en/procedures/28/28-018-017-tr.html>)
- Operate the engine at idle speed and inspect the engine for leaks.